

STUDY PROJECT – PHASE 1

DOCUMENT (PROBLEM IDENTIFICATION AND PLANNING)

Course Title: Study Project – Android Development

Project Title: Rydr – A Community-Driven Scenic Routing & Social Platform for Moto Riders

Student Name: Navneet, Debashis Maharana, Gowtham Sai G

Student ID: 2023EBCS685, 2023EBCS796, 2023EBCS761

Project Advisor / Supervisor: Swapnil Swarav

Date of Submission: 02,February 2026

1. PROJECT IDEA SUMMARY

Title of Project:

Rydr – A Community-Driven Scenic Routing & Social Platform for Moto Riders

Abstract:

Rydr is a community-centric mobile application designed for motorcycle riders, with future extensibility to car drivers and cyclists. The platform combines **scenic route recommendations (like Google Maps)**, **community discussions (like Reddit)**, and **social sharing (like Instagram/Twitter)** into a single ecosystem built specifically for road travelers.

Unlike traditional navigation apps that prioritize shortest or fastest routes, RideScape focuses on **ride quality**—scenic views, road conditions, curves, traffic density, and overall riding experience. Riders can discover and share the best routes for road trips, weekend rides, and long journeys.

The app also functions as a **Strava-like community for motor riders**, enabling users to track rides, log milestones, share stories, photos, and videos, participate in discussions, and compete on leaderboards. Rydr aims to become the digital home for riders—where navigation, storytelling, performance tracking, and community meet.

2. PROJECT BACKGROUND AND MOTIVATION

Problem Statement:

Motorcycle riders often value the **journey more than the destination**, yet existing platforms fail to serve this mindset:

- Google Maps optimizes for time, not experience
- Social media platforms lack structured route data
- Riding communities are scattered across WhatsApp groups, Instagram, Reddit, and forums
- No single platform exists to **discover scenic routes, discuss road conditions, and track riding achievements**

This fragmentation results in poor ride planning, lost local knowledge, and weak community engagement.

Motivation:

The growth of motorcycling culture, weekend road trips, and solo riding has created demand for a platform that:

- Prioritizes **scenic and enjoyable routes**
- Encourages **storytelling and knowledge sharing**
- Builds a **verified rider community**
- Supports long-term rider engagement beyond navigation

Academically, this project allows practical application of **product design, requirement analysis, system planning, and community-driven platforms**.

3. EDUCATIONAL VALUE AND COURSE ALIGNMENT

Relevance to Course Objectives:

This project applies:

- Software Development Life Cycle (SDLC)
- User-centric product thinking
- Requirement engineering
- Scalable system planning
- Real-world problem identification

Learning Outcomes:

- Product requirement documentation (PRD)
- Community-based system design
- Location-based service planning
- Feature prioritization & MVP thinking
- Translating user pain points into product features

4. OBJECTIVES

Primary Objectives:

- Build a rider-focused platform for scenic route discovery
- Enable riders to share routes, stories, photos, and videos
- Create a discussion-driven community similar to forums
- Track rides, milestones, and performance

Secondary Objectives:

- Encourage responsible and exploratory riding

- Preserve local route knowledge through community data
- Enable competitive and social engagement via leaderboards
- Lay foundation for multi-vehicle expansion (cars, cycles)

5. RESEARCH AND ANALYSIS

Existing Solutions:

Platform	Strength	Limitation
Google Maps	Accurate navigation	No scenic prioritization
Strava	Performance tracking	Limited social storytelling
Instagram	Visual sharing	No route intelligence
Reddit	Discussions	No location or ride context

Insight: No product unifies **routes + community + performance + storytelling** for riders.

Functional Requirements:

User & Community

- Rider profile with bike details
- Follow riders, like & comment on posts
- Discussion threads (forum-style)

Route & Navigation

- Scenic route recommendation engine
- Route tagging (mountain, coastal, twisty, highways)
- Community-rated routes

Content & Social

- Photo & video uploads
- Ride stories & travel logs
- Hashtags & location tagging

Tracking & Gamification

- Ride distance, duration, elevation
- Milestones & badges
- Leaderboards (region-based)

Non-Functional Requirements:

- **Usability:** Rider-friendly UI (glove-friendly, minimal taps)
- **Performance:** Real-time map & tracking
- **Scalability:** City → country → global expansion
- **Security:** User data protection & abuse moderation
- **Reliability:** Accurate GPS & offline support (future)

Feasibility Analysis:

- Technically feasible using modern mobile & map APIs
- MVP achievable within academic timeline
- Community-driven data reduces dependency on manual curation

6. PROJECT SCOPE AND EXPECTED DELIVERABLES

Scope Definition: Phase 1

Included:

- Rider profiles & social feed
- Basic scenic route discovery
- Ride logging & milestones
- Community discussions

Excluded:

- Turn-by-turn navigation replacement
- Monetization & subscriptions
- Brand partnerships & ads
- Offline maps

Deliverables

- Problem definition & research
- Functional & non-functional requirements
- High-level system architecture
- User flow & feature breakdown
- Phase-1 PRD documentation

7. PRELIMINARY PROJECT TIMELINE AND MILESTONES

Week 1–2: Problem identification and background research

Week 3: Requirement analysis and scope definition

Week 4: Phase 1 documentation and review

8. TEAM STRUCTURE AND COLLABORATION

Roles

- Product & Requirement Analysis
- System Design & Documentation
- UX Thinking & User Flow Mapping

Tools

- GitHub (version control)
- Figma (UX wireframes – optional)
- Google Docs (documentation)

9. RISK AND CHALLENGE ANALYSIS

Risks

- Low initial community adoption
- Inaccurate or biased route ratings
- Content moderation challenges
- GPS data inconsistencies

Mitigation Strategies

- Seed routes manually (Phase 1)
- Community moderation & reporting
- Reputation-based contributions

- Gradual regional rollout

10. SUPERVISOR REVIEW AND APPROVAL

Advisor Feedback:

Supervisor Comments:

Recommendations:

Signature:

Date: